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**THE IMPACT OF INTERNET SERVICE DOWNTIME ON THE
PROVISION OF HEALTH SERVICES THROUGH THE GOTHOMIS
SYSTEM AT PRIMARY HEALTH FACILITIES.**

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1.0 ABSTRACT

Electronic Health Record (EHR) systems are central to improving efficiency, continuity of care, and data quality in modern health systems; however, their effectiveness in low-resource settings is highly dependent on the reliability of supporting digital infrastructure. In Tanzania, the Government of Tanzania Hospital Management Information System (GoT-HoMIS) has been implemented nationwide to standardize clinical, administrative, and reporting processes across public health facilities, including lower-level facilities that form the foundation of primary healthcare delivery. Despite strong policy commitment, persistent internet service downtime continues to undermine the effective use of GoT-HoMIS, particularly in rural districts where ICT infrastructure, power stability, and technical support are limited. While national strategies acknowledge connectivity challenges, there remains a significant empirical gap at the district level regarding how internet service downtime directly affects routine health service delivery processes. This study therefore examines the impact of internet service downtime on the provision of health services through GoT-HoMIS in lower-level health facilities in Kwimba District Council, Tanzania. A cross-sectional mixed-methods design will be employed. Quantitative data will be collected through structured facility questionnaires administered to GoT-HoMIS users to assess outage frequency, duration, and associated service delivery disruptions. Qualitative data will be generated through key informant interviews with facility in-charges, clinicians, and data personnel to capture experiences, coping strategies, and workflow adaptations during system downtime. Observation and system log reviews will complement self-reported data. Quantitative data will be analysed using descriptive statistics, while qualitative data will be analysed thematically. By focusing on lower-level health facilities in a predominantly rural district, the study fills a critical practical and scientific gap by linking internet reliability to frontline health service delivery outcomes. The findings are expected to inform district and national decision-makers on infrastructure investment, contingency planning, and digital health system resilience in low-resource settings.

2.0 INTRODUCTION

2.1 Background of the study

Electronic Health Record (EHR) systems are globally recognized as a cornerstone of modern health systems due to their capacity to improve efficiency, continuity of care, data accuracy, and evidence-based clinical and managerial decision-making. Empirical evidence from high-income countries demonstrates that well-functioning EHR systems enhance clinical workflow, reduce medical errors, improve patient safety, and support population-level health planning (Boonstra et al., 2021; Adler-Milstein & Huckman, 2019).

In countries such as the United States, the United Kingdom, and across the European Union, EHR adoption has been institutionalized through strong digital infrastructure, reliable and redundant internet connectivity, comprehensive regulatory frameworks, and sustained public investment (OECD, 2020). Within these contexts, uninterrupted internet access is treated as a critical health system input on par with electricity, medicines, and skilled personnel because system downtime has been empirically linked to service delays, data loss, workflow disruption, and compromised patient safety (Ratwani et al., 2018).

At the regional level, sub-Saharan Africa has experienced a rapid expansion of digital health initiatives, driven by donor funding, national health reforms, and the pursuit of Universal Health Coverage (UHC). Countries such as Kenya, Rwanda, and Ghana have implemented national health information systems to improve service delivery, accountability, and reporting (Agarwal et al., 2020). However, regional empirical studies consistently demonstrate that internet instability remains one of the most persistent barriers to effective system utilization, particularly in rural and lower-level health facilities (Chukwu et al., 2021).

A multi-country assessment by the World Bank (2021) reported that over 60% of rural health facilities in sub-Saharan Africa experience frequent internet disruptions, significantly affecting patient registration, laboratory reporting, referral coordination, and continuity of care. These challenges are most pronounced at dispensary and health centre levels, despite their role as the primary entry point to the health system for most of the population.

Despite the demonstrated benefits of EHR systems, global evidence reveals a stark digital divide between high-income countries and low- and middle-income countries (LMICs). While LMICs have increasingly adopted digital health systems, their implementation and sustainability are frequently undermined by infrastructural deficits, particularly unreliable internet connectivity, unstable electricity supply, limited technical support, and inadequate user capacity (World Health Organization [WHO], 2019; Scott et al., 2020).

A systematic review by Fraser et al. (2022) found that internet service downtime in LMIC health facilities results in service interruptions, forced parallel use of paper-based systems, duplication of clinical and administrative work, increased risk of data errors, and deterioration in data quality. These findings underscore a critical limitation in existing digital health deployments: the effectiveness of EHR systems is not determined solely by software availability or policy endorsement, but by the reliability and resilience of the underlying digital infrastructure.

In Tanzania, the government has made deliberate policy commitments to digital transformation in the health sector. The Tanzania Digital Health Strategy 2019–2024 emphasizes the use of digital technologies to enhance efficiency, quality, and equity in health service delivery (Ministry of Health [MoH], 2019). Within this framework, the Government of Tanzania Hospital Management Information System (GoT-HoMIS) was introduced as a national EHR system intended to standardize clinical, administrative, and reporting functions across public health facilities.

However, despite its strategic importance, both policy documents and empirical evaluations explicitly acknowledge substantial implementation challenges. The Digital Health Strategy identifies “inadequate ICT infrastructure,” “unreliable internet connectivity,” and “unstable power supply” as major barriers to effective digital health implementation (MoH, 2019). Similarly, the Health Sector Strategic Plan V (2021–2026) notes that rural and lower-level health facilities face persistent connectivity challenges, limiting their ability to fully utilize electronic systems for service delivery and reporting (MoH, 2021).

Empirical studies conducted within Tanzania reinforce these concerns. Facility-level evaluations of GoT-HoMIS indicate that system usability and effectiveness are strongly dependent on the availability of stable internet connectivity. Mwogosi and Mambile (2023) found that although GoT-HoMIS improved record management and reporting accuracy in principle, frequent internet outages forced health workers to revert to manual documentation, resulting in delayed services, increased workload, and incomplete data entry.

Similarly, a case study in Meru District Hospital demonstrated that facilities with reliable local networks and consistent connectivity reported smoother GoT-HoMIS operations, while those experiencing frequent downtimes faced workflow disruptions, service delays, and staff frustration (Mtebe et al., 2022). These studies reveal a critical gap: while GoT-HoMIS exists as a national solution, its effectiveness is uneven and highly contingent on internet reliability.

Recent national events further expose this vulnerability. During the nationwide internet outage in mid-2024, caused by a submarine cable failure, health facilities across Tanzania reported disruptions in patient registration, laboratory result retrieval, and referral coordination. Media and stakeholder reports documented increased patient waiting times and delays in accessing critical health information (The Citizen, 2024). These events highlight the systemic risk posed by internet dependency in health service delivery, particularly in settings without redundancy or contingency mechanisms.

Within Kwimba District Council, these challenges are likely to be particularly pronounced. Kwimba is predominantly rural, and its health system is composed largely of dispensaries and health centres that operate under constrained infrastructural, financial, and human resource conditions. These lower-level facilities are mandated to use GoT-HoMIS for patient management and reporting; however, they often lack reliable internet connectivity, backup power sources, and on-site ICT support. As a result, internet service downtime may disproportionately disrupt health service provision at this level, where alternative systems are limited and patient volumes are high relative to available resources.

Despite national and regional evidence highlighting connectivity challenges, there remains a clear empirical gap at the district level regarding how internet service downtime specifically affects health service delivery through GoT-HoMIS in lower-level health facilities. Existing studies have largely focused on system adoption, usability, or general infrastructural barriers, with limited attention to quantifying the operational consequences of internet downtime on patient care processes.

This study therefore seeks to occupy this niche by systematically examining the impact of internet service downtime on the provision of health services through GoT-HoMIS in lower-level health facilities in Kwimba District Council. By linking connectivity disruptions to service delivery outcomes, the study directly responds to policy-identified gaps and provides evidence necessary for strengthening digital health system resilience at district and national levels.

2.2 Statement of the Problem

The Government of Tanzania has invested substantially in digital health systems as a strategic mechanism for improving efficiency, accountability, and quality of health service delivery, particularly through the nationwide implementation of the Government of Tanzania Hospital Management Information System (GoT-HoMIS) (Ministry of Health [MoH], 2019; MoH, 2021). However, the effective functioning of GoT-HoMIS is fundamentally dependent on the availability of stable and continuous internet connectivity, as the system operates primarily as an online, network-dependent platform (World Health Organization [WHO], 2019; Katuu, 2020).

In rural districts such as Kwimba District Council, lower-level health facilities operate within contexts characterized by limited ICT infrastructure, unreliable power supply, and weak or unstable internet connectivity (World Bank, 2021; Scott et al., 2020). Empirical studies conducted in low- and middle-income countries consistently demonstrate that such infrastructural constraints are associated with delays in patient registration, interruptions in clinical documentation, compromised data quality, increased workload for health workers, and reduced efficiency in service delivery (Fraser et al., 2022; Chukwu et al., 2021).

In Tanzania specifically, rural, and lower-level facilities have been shown to experience more frequent system interruptions compared to urban hospitals, largely due to poor broadband coverage and lack of redundancy in connectivity and power supply (Mtebe et al., 2022; Mwogosi & Mambile, 2023). Despite these challenges being acknowledged in national strategies and sector reports, they remain largely under-examined at the operational level of rural primary healthcare facilities (MoH, 2019; MoH, 2021).

Critically, while existing studies on GoT-HoMIS and other electronic health information systems in Tanzania have largely focused on system adoption, usability, and general infrastructural barriers, there is a notable lack of empirical, district-level evidence that systematically examines the direct impact of internet service downtime on the actual provision of health services (Katuu, 2020; Mtebe et al., 2022). In Kwimba District Council specifically, it remains unclear how frequently internet outages occur, how they disrupt day-to-day service delivery processes, and to what extent they affect the ability of lower-level health facilities to deliver timely and quality care through GoT-HoMIS.

Similar gaps have been noted in digital health research across sub-Saharan Africa, where studies often aggregate findings at national or regional levels, thereby masking important district- and facility-level variations (Agarwal et al., 2020; World Bank, 2021). The absence of localized empirical evidence limits the capacity of district health managers and national policymakers to design targeted interventions, prioritize infrastructure investments, and develop effective contingency mechanisms to enhance digital health system resilience (WHO, 2019; Scott et al., 2020).

Therefore, the central problem addressed by this study is that internet service downtime undermines the effective use of GoT-HoMIS, thereby disrupting the provision of health services in lower-level health facilities in Kwimba District Council. Without a clear understanding of the magnitude, nature, and consequences of these disruptions, efforts to strengthen digital health systems risk remaining policy-driven rather than evidence-informed (Petter et al., 2020; Fraser et al., 2022). This study seeks to address this gap by empirically examining the impact of internet service downtime on health service delivery through GoT-HoMIS in lower-level health facilities in Kwimba District Council, Tanzania, thereby generating evidence to inform both district-level management decisions and national digital health policy implementation.

2.3 Objectives of the study

2.3.1 General objective

To examine the impact of internet service downtime on the provision of health services through the GoTHoMIS system in lower level health facilities in Tanzania. A case of Kwimba DC Health Facilities

2.3.2 Specific objectives

- i. To assess the frequency of Internet outages affecting GoT-HoMIS system in lower level health facilities
- ii. To identify the effects of Internet outages on health service delivery processes in lower level health facilities
- iii. Explore coping strategies employed by healthcare workers' during GoT-HoMIS system failure in lower level health facilities

2.4 Research questions

- i. How frequent are Internet outages affecting GoT-HoMIS system in lower level health facilities?
- ii. What are the effects of Internet outages on health service delivery processes in lower level health facilities?
- iii. What are the coping strategies employed by healthcare workers' during GoT-HoMIS system failure in lower level health facilities?

2.5 Justification of the Study

The justification for this study is both practical and policy-relevant.

From a practical perspective, lower-level health facilities serve as the backbone of Tanzania's primary healthcare system. Understanding how internet downtime affects service provision at this level is essential for improving patient care, staff efficiency, and data integrity.

From a policy perspective, the study responds directly to gaps identified in national digital health strategies, which call for evidence-based interventions to strengthen ICT infrastructure and system resilience. By generating empirical evidence from Kwimba DC, this study will inform district planners, national policymakers, and development partners on how to align digital health investments with on-the-ground realities, thereby enhancing the effectiveness and sustainability of GoT-HoMIS and similar systems.

2.6 Limitations, Assumptions, and Constraints

This study is subject to several limitations.

First, the cross-sectional design captures conditions at a single point in time and does not account for seasonal or long-term variations in internet reliability.

Second, reliance on self-reported data may introduce recall bias, particularly regarding the frequency and duration of internet outages.

Third, system downtime logs may be incomplete or inconsistently maintained across facilities. The study assumes that respondents provide honest and accurate information and that GoT-HoMIS usage patterns during the study period are representative of routine operations.

Financial and time constraints limit the scope of the study to one district, which may affect broader generalization; however, the findings remain analytically relevant to similar rural districts in Tanzania.

3.0 LITERATURE REVIEW

3.1 Theoretical Foundations of Digital Health Systems

3.1.1 Technology–Organization–Environment (TOE) Framework

The Technology–Organization–Environment (TOE) framework was originally developed by Tornatzky and Fleischer in 1990 as part of their work on the processes of technological innovation within organizations. The framework was designed to explain how organizations adopt, implement, and utilize technological innovations by considering three interrelated contexts: the technological context, the organizational context, and the environmental context. Since its development, the TOE framework has been widely applied and empirically validated across multiple sectors, including manufacturing, banking, education, and, more recently, healthcare, and digital health systems (Oliveira & Martins, 2011).

The technological context refers to both internal and external technologies relevant to the organization, including infrastructure availability, compatibility, complexity, and reliability. In the context of GoT-HoMIS, this includes internet connectivity, network stability, system uptime, availability of hardware, and power supply. Internet service downtime directly undermines technological readiness, rendering GoT-HoMIS partially or fully unusable and disrupting service delivery.

The organizational context encompasses characteristics of the organization such as size, human resource capacity, managerial support, ICT skills, workflow design, and internal processes. In lower-level health facilities in Tanzania, organizational capacity is often constrained by limited staffing, inadequate ICT training, and high workload. When internet downtime occurs, these organizational weaknesses amplify its impact, as health workers may lack the skills or protocols to manage system interruptions effectively.

The environmental context includes external factors such as national policies, regulatory frameworks, donor support, ICT infrastructure coverage, and relationships with service providers. In Tanzania, environmental factors include national digital health policies, the reliability of internet service providers, and the broader rural ICT ecosystem. Weak rural broadband infrastructure and limited redundancy in internet services expose lower-level facilities to frequent downtimes beyond their control.

The TOE framework is particularly strong for this study because it explicitly recognizes that system performance is influenced not only by the technology itself but also by organizational capacity and environmental conditions. It allows the study to systematically analyse how internet service downtime (technological factor), facility preparedness and staff capacity (organizational factors), and rural ICT infrastructure and policy context (environmental factors) interact to affect GoT-HoMIS performance and health service delivery.

Despite its strengths, the TOE framework has limitations. It is primarily an adoption-oriented model and does not explicitly address user behaviour, workflow adaptation, or the dynamic interactions between humans and technology during system failure. In the context of this study, TOE alone may not fully explain how health workers cope with downtime through informal workarounds or how such adaptations affect data quality and service outcomes. This limitation necessitates complementing TOE with socio-technical perspectives.

3.1.2 Socio-Technical Systems Theory

Socio-Technical Systems Theory originated from the work of researchers at the Tavistock Institute in the United Kingdom during the 1950s, particularly Eric Trist and Ken Bamforth (1951). The theory was developed in response to observations that technological changes in organizations often failed when human and social dimensions were ignored. The central premise of the theory is that organizations are composed of interdependent social and technical subsystems, and optimal performance is achieved only when both are jointly optimized.

The technical subsystem includes tools, technologies, infrastructure, and processes required to perform tasks. In the case of GoT-HoMIS, this encompasses the software platform, internet connectivity, servers, power supply, and hardware devices. The social subsystem includes people, skills, roles, communication patterns, organizational culture, and workflows. In lower-level health facilities, this involves health workers' competencies, routines, workload distribution, and informal practices.

According to Socio-Technical Systems Theory, digital health systems function effectively only when there is alignment between these two subsystems. Internet service downtime disrupts this alignment by rendering the technical system unreliable, forcing health workers to modify workflows, rely on memory or paper records, and engage in informal practices to

maintain service delivery. While these workarounds allow services to continue, they often compromise efficiency, data accuracy, and continuity of care.

This theory is highly relevant to the study because it explains how internet downtime affects not only system functionality but also human behaviour and organizational processes. It provides a strong lens for understanding how health workers adapt to GoT-HoMIS disruptions, why parallel paper systems persist, and how these adaptations influence service quality. It is particularly useful for interpreting qualitative findings related to coping strategies and workflow disruptions.

A key limitation of Socio-Technical Systems Theory is that it offers limited guidance on macro-level factors such as policy, regulatory environments, and national infrastructure. While it explains human–technology interactions well, it does not explicitly account for broader environmental determinants such as rural ICT coverage or national digital health governance. Therefore, while valuable, it requires integration with frameworks like TOE to provide a comprehensive explanation.

3.1.3 Information Systems Success Model

The Information Systems Success Model was originally proposed by William H. DeLone and Ephraim R. McLean in 1992 and later refined in 2003 in response to extensive empirical testing and critique. The model was designed to assess the effectiveness and success of information systems by identifying key dimensions that influence system outcomes. The updated model identifies six interrelated constructs: system quality, information quality, service quality, system use, user satisfaction, and net benefits.

System quality refers to the technical performance of the system, including reliability, availability, response time, and ease of use. Internet reliability is a core component of system quality in web-based health information systems such as GoT-HoMIS. Frequent internet service downtime directly reduces system availability and responsiveness, thereby lowering perceived system quality.

Information quality relates to the accuracy, completeness, timeliness, and relevance of information produced by the system. During internet downtime, data entry is often delayed or performed retrospectively, increasing the risk of incomplete or inaccurate records. Service quality refers to the quality of technical and user support provided to system users, which is often limited in lower-level health facilities. According to the model, poor system quality

leads to reduced system use and user dissatisfaction, ultimately diminishing the net benefits of the system, such as improved service efficiency and patient outcomes (Petter et al., 2020).

The Information Systems Success Model is particularly strong in linking internet reliability to measurable outcomes such as user satisfaction, system usage, and service delivery benefits. It provides a clear pathway for examining how internet downtime affects GoT-HoMIS performance and health service outcomes, making it well suited for the quantitative component of the study. The model assumes that users have discretion over system use, which may not fully apply in mandated systems like GoT-HoMIS. Additionally, it places less emphasis on contextual and environmental constraints, such as rural infrastructure limitations, which are central to this study. As such, while the model is valuable for assessing system outcomes, it must be contextualized within broader organizational and environmental realities.

3.2 Empirical literature review

Empirical studies from high-income countries demonstrate that stable internet connectivity significantly enhances EHR performance and service delivery outcomes.

A longitudinal study in the United States found that facilities with near-zero system downtime achieved faster patient throughput, improved clinical documentation accuracy, and better coordination of care (Adler-Milstein et al., 2019). Similarly, a UK-based study reported that even short EHR outages resulted in increased waiting times and elevated clinical risk, prompting the NHS to mandate redundancy and downtime protocols (Greenhalgh et al., 2020). However, these studies are conducted in contexts characterized by advanced infrastructure and strong institutional support. Their findings, while robust, are not directly transferable to low-resource settings where infrastructure reliability cannot be assumed. This limitation highlights the need for context-specific research in LMICs.

In LMICs, empirical literature consistently identifies unreliable internet connectivity as a major constraint to effective digital health implementation. A systematic review by Scott et al. (2020) covering digital health initiatives across Africa and Asia found that over 70% of failed or underperforming EHR systems cited connectivity issues as a primary factor. Fraser et al. (2022) further demonstrated that internet service downtime leads to workflow fragmentation, data duplication, and loss of clinical continuity. Studies conducted in countries such as India, Nigeria, and Bangladesh reveal that health workers often maintain parallel paper systems

during outages, undermining data completeness and increasing workload (Chukwu et al., 2021; Khan et al., 2023).

While these studies provide valuable insights, many rely heavily on qualitative data and self-reported experiences, with limited measurement of actual service delivery impacts such as patient waiting time or service interruption duration.

In sub-Saharan Africa, digital health expansion has outpaced infrastructure development in many rural areas. A World Bank (2021) assessment reported that frequent internet outages disrupted core health service processes in rural facilities, including patient registration, laboratory reporting, and referral coordination. Empirical studies from Kenya and Uganda similarly found that system downtime directly affected patient flow and health worker productivity (Were et al., 2020; Nabukenya et al., 2022). However, most regional studies focus on system adoption and user perceptions rather than operational service delivery outcomes. Furthermore, they often aggregate data at national or regional levels, masking district-level variations and facility-specific challenges.

Tanzania has generated a growing body of literature on digital health systems, particularly GoT-HoMIS. Mwogosi and Mambile (2023) found that GoT-HoMIS improved reporting accuracy but was highly vulnerable to internet instability, leading to frequent reversion to paper-based systems. Mtebe et al. (2022) similarly reported that inadequate network infrastructure disrupted clinical workflows and reduced system utilization. While these studies provide important insights, they exhibit several limitations. First, most focus on system usability and user satisfaction rather than the direct impact of internet downtime on service delivery outcomes. Second, many studies are conducted in hospitals or urban settings, leaving lower-level rural facilities under-represented. Third, few studies integrate both quantitative and qualitative methods to triangulate outage frequency with service disruption effects.

Policy documents such as the Tanzania Digital Health Strategy (2019–2024) and the Health Sector Strategic Plan V (2021–2026) acknowledge internet unreliability as a challenge but do not provide empirical district-level data to guide targeted interventions. This gap between policy recognition and empirical evidence underscores the need for localized studies. This study positions itself within the existing literature by focusing explicitly on internet service downtime as a determinant of health service delivery performance, rather than merely system

adoption or user perception. By adopting a mixed-methods approach and concentrating on lower-level health facilities in Kwimba District Council, the study responds directly to identified gaps in both empirical and policy-oriented literature.

3.3 Conceptual Framework of the Study

The conceptual framework guiding this study is informed by the Technology–Organization–Environment (TOE) framework, Socio-Technical Systems Theory, and the Information Systems Success Model. The framework conceptualizes internet service downtime as the key independent variable influencing health service delivery outcomes through GoT-HoMIS. Internet downtime is operationalized in terms of frequency, duration, and predictability. The relationship between internet downtime and service delivery outcomes is mediated by organizational and environmental factors, including availability of backup internet and power, ICT skills of health workers, and facility preparedness and contingency mechanisms. Health service delivery outcomes constitute the dependent variables and are reflected in patient waiting time, service continuity, timeliness of records, and workflow efficiency.

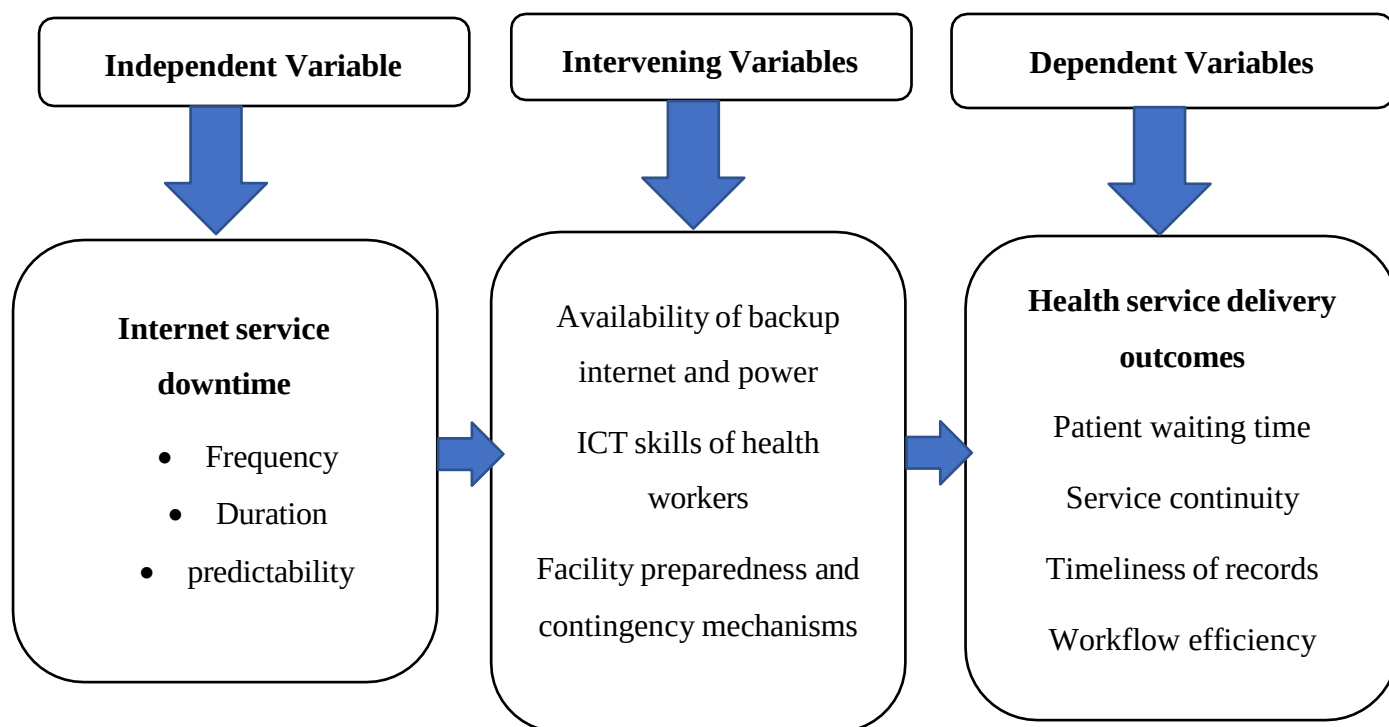


Figure 1: Conceptual Framework of the Study

Source: Researchers Own Construct (2025)

4.0 RESEARCH PLAN AND METHODOLOGY

4.1 Study Design

This study will adopt a cross-sectional research design, integrating both quantitative and qualitative approaches through concurrent data collection. The design is particularly appropriate for examining complex health systems phenomena, such as the impact of internet service downtime on digital health service delivery, where both measurable outcomes and human experiences are critical to comprehensive understanding.

A cross-sectional approach is appropriate because the study seeks to examine existing conditions and relationships at a specific point in time rather than longitudinal changes. This design is well suited to resource-constrained settings and has been widely applied in health systems and digital health research to assess system performance and service delivery outcomes.

4.2 Study Area and Study Population

The study will be conducted in Kwimba District Council, located in Mwanza Region, Tanzania. Kwimba District is predominantly rural, characterized by dispersed settlements and limited ICT infrastructure. The district health system comprises one district hospital, several health centres, and numerous dispensaries, which together form the primary point of healthcare access for most of the population.

The focus of this study will be on lower-level health facilities, specifically health centres and dispensaries, as these facilities are mandated to use the Government of Tanzania Hospital Management Information System (GoT-HoMIS) while simultaneously facing the most pronounced infrastructural constraints. Selected facilities will include health centres such as Budushi and Isunga, and dispensaries such as Bugandando and Bugembe, among others.

The target population will consist of healthcare workers directly involved in the use, management, or supervision of GoT-HoMIS. These include clinicians, nurses, data clerks, health records officers, and facility in-charges. These cadres are best positioned to provide accurate information on system functionality, service delivery disruptions, and operational responses during internet service downtime.

4.3 Sampling Procedures and Sample Size

The study will employ a stratified purposive sampling strategy to ensure representation across different facility types and geographical areas within the district. All health centres in Kwimba District Council will be included due to their limited number and their strategic role in service delivery. In addition, approximately 15 dispensaries will be purposively selected to represent different wards, population catchment sizes, and levels of infrastructural accessibility. The district hospital will be included primarily for contextual comparison rather than as a central unit of analysis.

Within each selected facility, all eligible GoT-HoMIS users present during the data collection period will be invited to participate in the quantitative survey. For the qualitative component, one to two key informants per facility typically the facility in-charge, a senior clinician, or a data officer will be purposively selected based on their roles and experience with GoT-HoMIS operations. The study anticipates collecting approximately 100 completed questionnaires and conducting 20–30 in-depth interviews. This sample size is considered adequate for identifying meaningful patterns and relationships within a district-level study while allowing for data saturation in the qualitative component.

4.4 Data Collection Methods

4.4.1 Facility Questionnaire

A structured questionnaire will be administered to GoT-HoMIS users at each sampled facility. The instrument will include sections on internet service downtime, service delivery impacts, and facility readiness. Items will be mapped directly to research objectives and adapted from validated HMIS instruments. Pre-testing will be conducted to refine clarity and relevance.

4.4.2 Key Informant Interview Guide

A semi-structured interview guide will be used to elicit detailed qualitative insights from facility staff. The guide will be reviewed by subject experts and piloted prior to full implementation. Interviews will be conducted consistently to ensure reliability while allowing depth of exploration.

4.4.3 Observation and Log Review Checklist

A standardized checklist will be used during facility visits to document infrastructural and procedural characteristics, including backup power availability, alternative internet sources, GoT-HoMIS workstations, and use of paper-based records during downtime. This instrument will support validation of self-reported data and provide objective contextual information

4.5 Data Analysis

4.5.1 Quantitative Data Analysis

Quantitative data will be entered and analysed using statistical software such as SPSS or Stata. Descriptive statistics, including frequencies, means, medians, and standard deviations, will be used to summarize patterns of internet downtime and service delivery indicators across facilities.

4.5.2 Qualitative Data Analysis

Qualitative data will be analysed using thematic analysis. Transcripts will be coded inductively and deductively to identify recurring patterns and themes related to service disruption, workflow adaptation, and system resilience. Coding will be conducted iteratively, with emerging themes refined through comparison across interviews.

To enhance analytical rigor, multiple researchers will independently code selected transcripts and reconcile differences through discussion. Qualitative data will be managed using software such as NVivo to ensure systematic organization and transparency. Findings from the qualitative analysis will be used to contextualize and explain quantitative results.

4.6 Validity, Reliability, and Trustworthiness

To enhance validity, the study will employ methodological triangulation by integrating survey data, interview findings, and observational or log-based evidence. Research instruments will be informed by established HMIS assessment tools and relevant literature to ensure content validity.

Reliability will be strengthened through standardized data collection procedures, including enumerator training and consistent use of research instruments. Internal consistency of multi-item scales will be assessed using Cronbach's alpha, with a threshold of $\alpha \geq 0.7$ considered acceptable. For qualitative data, inter-coder agreement and audit trails will enhance dependability.

While the findings will be context-specific to Kwimba District Council, the inclusion of diverse facility types enhances analytical generalizability to similar rural districts in Tanzania. Limitations such as recall bias, incomplete system logs, and cross-sectional design constraints will be explicitly acknowledged and discussed.

Content validity of the research instruments will be ensured through expert review by digital health and research methodology specialists. Construct validity will be supported by aligning questionnaire items directly with variables defined in the conceptual framework. Reliability of the quantitative instrument will be assessed using Cronbach's alpha for multi-item scales, with a minimum threshold of 0.7 considered acceptable. Qualitative trustworthiness will be enhanced through triangulation, consistent interviewing procedures, and audit trails.

5.0 Research Instruments

Annex I: Facility Questionnaire (Quantitative Survey)

Title: Facility Questionnaire on Internet Service Downtime and Health Service Delivery through GoT-HoMIS

Target Respondents: Clinicians, nurses, data clerks, health records officers, and facility in-charges

Instruction: Please tick (✓) or write the most appropriate response. All information will be treated with strict confidentiality and used for academic purposes only.

Section A: Respondent and Facility Characteristics

1. Facility name: _____
2. Facility type:
 - ☐ Health Centre
 - ☐ Dispensary
3. Respondent cadre:
 - ☐ Clinician
 - ☐ Nurse
 - ☐ Data Clerk / Health Records Officer

☐ Facility In-charge

☐ Other (specify): _____

4. Years of experience using GoT-HoMIS:

☐ Less than 1 year

☐ 1–3 years

☐ 4–6 years

☐ More than 6 years

5. Average number of patients served per day:

☐ Less than 30

☐ 30–60

☐ 61–100

☐ More than 100

Section B: Internet Connectivity and Downtime

6. How often has your facility experienced internet service downtime in the last 3 months?

☐ Never

☐ 1–2 times

☐ 3–5 times

☐ More than 5 times

7. What is the average duration of internet downtime?

☐ Less than 30 minutes

☐ 30 minutes–2 hours

☐ 2–6 hours

☐ More than 6 hours

8. Internet downtime usually occurs during:

☐ Working hours

☐ Non-working hours

☐ Both

9. Are you usually informed in advance about internet interruptions?

☐ Yes

☐ No

Section C: GoT-HoMIS System Availability and Use

10. During internet downtime, GoT-HoMIS becomes:

☐ Completely inaccessible

☐ Partially accessible

☐ Fully accessible

11. How often do you enter patient data offline due to internet downtime?

☐ Never

☐ Sometimes

☐ Often

☐ Always

12. How long does it take to synchronize offline data after internet restoration?

☐ Immediately

☐ Within the same day

☐ 1–3 days

☐ More than 3 days

Section D: Impact on Health Service Delivery

13. During internet downtime, patient waiting time:

- ☐ Does not change
- ☐ Slightly increases
- ☐ Moderately increases
- ☐ Significantly increases

14. Internet downtime leads to delays in: (Tick all that apply)

- ☐ Patient registration
- ☐ Clinical documentation
- ☐ Laboratory results
- ☐ Referrals
- ☐ Reporting

15. Have services ever been postponed due to GoT-HoMIS unavailability?

- ☐ Yes
- ☐ No

Section E: Organizational and Environmental Factors

16. Does the facility have backup internet (e.g., mobile data)?

- ☐ Yes
- ☐ No

17. Does the facility have backup power supply?

- ☐ Generator
- ☐ UPS
- ☐ None

18. Is there an ICT support person available when systems fail?

☐ Yes

☐ No

Section F: Perceptions (Likert Scale)

(1 = Strongly Disagree ... 5 = Strongly Agree)

19. Internet downtime significantly delays patient care.

20. Manual documentation during downtime increases staff workload.

21. GoT-HoMIS improves service delivery when internet is stable.

22. Internet reliability is critical to effective GoT-HoMIS use.

Annex II: Key Informant Interview Guide (Qualitative)

Title: Interview Guide on Internet Service Downtime and GoT-HoMIS Use

Target Participants: Facility in-charges, senior clinicians, data officers

Instruction: The interviewer should probe for examples and clarification.

Section A: Background Information

1. Please describe your role and responsibilities in this facility.
2. How long have you been using GoT-HoMIS?

Section B: Experience with Internet Downtime

3. Can you describe a recent incident when internet services were unavailable?
4. How frequently do such internet interruptions occur in this facility?

Section C: Impact on Service Delivery

5. What happens to patient registration and clinical documentation during internet downtime?
6. How does downtime affect patient waiting time and service continuity?
7. Are there services that cannot be provided when GoT-HoMIS is inaccessible?

Section D: Socio-Technical Interactions

8. How do health workers adjust their workflow during system downtime?
9. What challenges arise when shifting from electronic to manual processes?

Section E: Coping Mechanisms and Support

10. What strategies does the facility use to cope with internet downtime?
11. What kind of support do you receive from the district or higher levels?

Section F: Recommendations

12. What measures would you recommend to reduce the impact of internet downtime on service delivery?

Annex III: Observation and Log Review Checklist

Title: Facility Observation and Log Review Checklist

Completed by: Researcher during facility visit

SECTION A: INFRASTRUCTURE AVAILABILITY

Item	Yes	No
Stable electricity supply	☐	☐
Backup power (generator/UPS)	☐	☐
Primary internet connection available	☐	☐
Alternative internet source available	☐	☐

Section B: GoT-HoMIS Readiness

Item	Yes	No
Functional GoT-HoMIS workstations	☐	☐
Staff trained on GoT-HoMIS	☐	☐
Written procedures for system downtime	☐	☐

Section C: Operational Practices During Downtime

Item	Yes	No
Paper registers used during downtime	☐	☐
Evidence of delayed data entry	☐	☐
Backlog of unsynchronized records	☐	☐

Section D: Log Review

- Presence of IT incident or downtime logs: ☐ Yes ☐ No
- DHIS2/GoT-HoMIS submission delays observed: ☐ Yes ☐ No
- Dates of observed outages (if recorded): _____

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